

AGNEY K RAJEEV

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ABOUT ME

I am an **Integrated Master of Science graduate** from **NISER Bhubaneswar**. I majored in **Physics** (with a minor in **Computer Science**) with a particular inquisitiveness for **Statistical Mechanics, Soft Condensed Matter Physics and Biological Physics**. My specific curiosity is in the **physics of living matter** which lies at the **intersection of Physics, Biology and Computational Science**. I am interested in understanding complex biological relations and systems such as bird flocks, bacterial colonies, proteins in media etc., creating interaction-based physics models for the same and using computer simulations (aided by machine learning protocols if necessary) to evolve, observe and analyse emergent behaviours.

RESEARCH EXPERIENCE

- **Computational Studies of The Ising Model** Jun 2023 - Aug 2023
NISER Bhubaneswar
Summer Internship
 - Guide: [Dr. A. V. Anil Kumar](#)
 - Studied the 2D Ising model using the Metropolis Algorithm
 - Determined the existence of a phase transition and evaluated the Critical Temperature
 - Attempted to incorporate non-reciprocal interactions and random fields to the model
- **Predicting Protein Oligomericity from Amino Acid Sequence** Jan 2024 - Sep 2024
NISER Bhubaneswar
Semester Project
 - Guide: [Dr. Subhankar Mishra](#)
 - Attempted to create a machine learning model for predicting the oligomerization states of proteins from amino acid sequences
 - Achieved near state-of-the-art results in classifying fluorescent proteins into monomers and oligomers
- **Predicting the Stress Tensor for a simulated molecular system** May 2024 - Jul 2024
IISER Kolkata
Summer Project
 - Guide: [Dr. Neelanjana Sengupta](#)
 - Sought to develop an efficient alternative to predicting Stress Tensor for biomolecules in a solution using deep learning
 - Successfully obtained novel preliminary results for the atomic stress tensor of a water box using node regression learning in Graph Neural Networks
- **Effects of non-reciprocal interactions in the Ising Universality class** Jun 2025 - May 2026
NISER Bhubaneswar
MSc. Thesis
 - Guide: [Dr. A. V. Anil Kumar](#)
 - Incorporated non-reciprocal couplings to the Ising ferromagnet and analyzed critical and dynamical behaviour
 - Integrated non-reciprocal dispersal to a metapopulation model in Ising universality
 - Attempt to observe any change in phase transitions, critical values or universality class
 - Also study the dynamic behaviour of the new model
 - repeated the same for the Non-equilibrium Random Field Ising Model(NRFIM)

EDUCATION

- **National Institute of Science Education and Research** 2021 - 2026
Bhubaneswar, India
Integrated MSc.
 - CGPA: 8.99/10.00
- **Chinmaya Vidyalaya Kolazhy** 2009 - 2021
Thrissur, India
Higher Secondary School
 - Class XII: 97.0/100.0
 - Class X: 95.0/100.0

PUBLICATIONS

C=CONFERENCE, P=PREPRINT

- [P.1] Agney K. Rajeev, & A. V. Anil Kumar. (2024). [Ising model with non-reciprocal interactions](#).
- [C.1] Rajeev, A.K., B, J.J.K. & Mishra, S.. (2025). [Predicting Oligomeric states of Fluorescent Proteins using Mamba](#). *Proceedings of the 6th Northern Lights Deep Learning Conference (NLDL)*, in *Proceedings of Machine Learning Research* 265:204-212.

SKILLS

- **Programming Languages:** Python, NetLogo, Julia, Fortran, Javascript
- **Data Science & Machine Learning:** Pandas, PyTorch
- **Mathematical & Statistical Tools:** MATLAB, Mathematica
- **Other Tools & Technologies:** Arduino, SeeLab

HONORS AND AWARDS

- **99.64 percentile** 2026
Council of Scientific and Industrial Research National Eligibility Test (CSIR-UGC NET)
 - National-level examination in India determining the eligibility of Indian nationals for a Junior Research Fellowship (JRF), appointment as an Assistant Professor, and admission to Ph.D. programs in major science disciplines.
- **Rank 528** 2025
Graduate Aptitude Test in Engineering (GATE)
 - National-level entrance exam in India for postgraduate admission to Master's and PhD programs in engineering, technology, science, architecture, and humanities.
- **Semester IV Topper** 2023
NISER Bhubaneswar
 - GPA: 9.36
- **Scholar** 2021
Innovation in Science Pursuit for Inspired Research (INSPIRE)
 - Innovative programme sponsored and managed by the Department of Science & Technology for attraction of talent to Science
 - Eligible for top 1% students in Class XII exams

RELEVANT COURSEWORK

Master's Coursework

- **Physics :** Classical Mechanics I & II, Mathematical Methods I & II, Quantum Mechanics I & II, Statistical Mechanics, Introduction to Condensed Matter Physics, NonEquilibrium Statistical Mechanics, NonLinear Dynamics and Chaos, Phase Transitions and Critical Phenomena, Computational Physics
- **Computer Science :** Introductory Python Programming I & II, Theory of Computation, Discrete Structures and Computation, Design and Analysis of Algorithms, Machine Learning, Complexity Theory, Parametrized algorithms
- **Biology :** Introductory BioPhysics, Evolutionary Biology

Self learning

- Probability Theory, Physics Informed Machine Learning, Evolutionary Game Theory, Agent-based Modeling(ongoing)

ADDITIONAL INFORMATION

Languages: Malayalam(Mother Tongue), English(Fluent), Hindi(Fluent), Spanish(Very Basic)

Interests: Table Tennis, Piano, Coding, Cryptography, Simulation games

REFEREES

Dr. A. V. Anil Kumar

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Dr. Subhankar Mishra

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